

EMOTION RECOGNITION SYSTEM

A PROJECT REPORT

Submitted by

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SRM INSTITUTE OF SCIENCE & TECHNOLOGY

(Under Section 3 of UGC Act, 1956)

BONAFIDE CERTIFICATE

Certified that this project report titled “**EMOTION RECOGNITION SYSTEM**” is the bonafide work of “**ARPIT KHARE [Reg No: RA1811003030377], ARCHISMAAN BANERJEE [Reg No: RA1811003030382], RAJASHVI SRIVASTAVA [Reg No: RA1811003030401], ADITI ANAND [Reg No: RA1811003030405]**”, who carried out the project work under my supervision. Certified further, that to the best of my knowledge the work reported herein does not form any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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ABSTRACT

In today's world, the face uses expressions to convey tons of knowledge visually, hence Emotion-Recognition System plays a major role within the space of computer-human interaction. Our facial emotions are unit conveyed through the stimulation of unique sets of face muscles. They are generally refined, nevertheless advanced, signals in an expression that always contains a verdant quantity of knowledge concerning our mindset. By emotion recognition (classification), We tend to design a deep learning neural network that offers computers the power to create speculation concerning sentiments. The most objective of our project is that we tend to applied numerous deep learning ways like convolutional neural networks to spot general human emotions.

ACKNOWLEDGEMENTS

We would like to express our deepest gratitude to our guide, Amit Sharma his valuable guidance, consistent encouragement, personal caring, timely help and providing me with an excellent atmosphere for doing research. All through the work, in spite of his busy schedule, he has extended cheerful and cordial support to us for completing this research work.

Aditi Anand

Rajashvi Srivastava

Archismaan Banerjee

Arpit Khare

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ABBREVIATIONS

FER - Facial Emotion Recognition

E.R.S.- Emotion Recognition System

A.I.- Artificial Intelligence

C.N.N.- Convolutional Neural Network

SDLC- Software Development life cycle

ANN - Artificial Neural Network

UML -Unified Modeling Language

RELU- Rectified Linear Units

CHAPTER 1

INTRODUCTION

Feeling acknowledgment assumes an essential part in the period of Artificial insight and the Internet of things. It offers a huge extension to human PC collaboration, mechanical technology, medical care, bio metric security, and social demonstration. Feeling acknowledgment frameworks perceive feelings from facial articulations, text information, body developments, voice, cerebrum, or heart signals. Alongside fundamental feelings, disposition, command over feelings, and force of initiation of feeling can likewise be inspected for examining opinions. This idea distinguishes different directed and unaided AI strategies for including extraction and feeling characterization. A similar investigation has likewise been made of different machine learning calculations utilized in referred to papers. It tells the extension and uses of programmed feeling acknowledgment frameworks in different fields. This idea additionally talks about different boundaries to expand the precision, security, and productivity of the framework.

1.1 The significant targets and thoughts include:

1. Tracing the human faces in the picture (i.e. this step is called as finding the face),
2. Selecting emotion features in the traced area in the frame (i.e. this step is called as detecting face attribute),
3. Examining the behaviour of face physiognomy, modifications, appearance of face and by classifying the familiarity in some special facial-expression categories like the distribution of emotions.

CHAPTER 2

LITERATURE SURVEY

Table 2.1: Literature Survey

Author Name	Title	Source	Findings
Byoung Chul Ko	A Brief Review of Facial Emotion Recognition Based on Visual Information	Department of Computer Engineering, Keimyung University, Daegu 42601, Korea; niceko@kmu.ac.kr;	The classification algorithms utilized in standard FER embody SVM, Adaboost, and random forest; by distinction, deep-learning-based FER approaches extremely cut back the dependence on face-physics-based models and different pre-processing techniques by sanctioning "end-to-end" learning within the pipeline directly from the input pictures.
Ronak Kosti, Jose M. Alvarez, Adria Recasens, Agata Lapedriza	Emotion Recognition in Context	Universitat Oberta de Catalunya Data61 / CSIRO Massachusetts Institute of Technology	They planned a CNN model for the task of feeling estimation in context. The model is based on state-of-the-art techniques for visual recognition and provides a benchmark on the proposed downside of estimating emotional states in context.
Shashidhar G. Koolagudi K. Sreenivasa Rao	Emotion recognition from speech: a review	© Springer Science+ Business Media, LLC 2011	researchers use non-linear classifiers forever at the value of complexness and procedure time. However systematic approach supported nature of speech features is needed while selecting the pattern classifiers for emotion recognition.

Bjorn Schuller, Gerhard Rigoll, and Manfred Lang	HIDDEN MARKOV MODEL-BASED SPEECH EMOTION RECOGNITION	Institute for Human- Computer Communication Technische Universitat Miinchen	The two introduced methods have proven each capable of a rather reasonable model for the automated recognition of human emotions in speech.
Alvin I. Goldman, Chandra Sekhar Sripada	Simulationist models of face-based	Center for Cognitive Science, Rutgers University, P.O. Box 1179, Piscataway, NJ 08855-1179, USA	Four different models have been explored: a generate-and-test model, a reverse simulation model, a variant of the reverse simulation model that employs an associate "as if" loop, associated an immediate resonance model.
Mostaz El Ayadi, Mohamed S. Kamel, Fakhri Karray b	Survey on speech emotion recognition: Features, classification schemes,	Engineering Mathematics and Physics, Cairo University, Giza 12613, Egypt	the performance of current stress detectors still desires vital improvement. The typical classification accuracy of speaker-independent speech emotion recognition systems are less than eightieth in most of the planned techniques.
Suraj Tripathi1, Abhay Kumar1*, Abhiram Ramesh1*, Chirag Singh1*, Promod	Deep Learning-based Emotion Recognition System Using	Samsung R&D Institute India – Bangalore	Speech options primarily based 2D CNN model provides better accuracy relative to progressive results, which more improves when combined with text.
Emily Mower, Student Member, IEEE, Maja J Mataric', Senior Member, IEEE, and Shrikanth Narayanan, Fellow, IEEE	A Framework for Automatic Human Emotion	IEEE TRANSACTIONS ON AUDIO, SPEECH, AND LANGUAGE PROCESSING, VOL. 19, NO. 5, JULY 2011	the utility of emotion- based parts for representation rather than data-driven clusters as the relevant parts in the profile construction. These analyses can offer further proof concerning the effectuality of the profile -based representations of feeling.

Ryoko TOKUHISA, Kentaro INUI, Yuji MATSUMOTO	Emotion Classification Using Massive Examples Extracted from the Web	Proceedings of the 22nd International Conference on Computational Linguistics (Coling 2008)	In this paper, we have addressed the problem of emotion classification assuming its potential applications to be human -computer dialog system as well as active-listening dialog
Roddy Cowie, Ellen Douglas-Cowie, Nicolas Tsapatsoulis, George Votsis, Stefanos Kollias, Winfried Fellenz, and John G Taylor.	Emotion Recognition in Human-Computer Interaction	IEEE SIGNAL PROCESSING MAGAZINE JANUARY 2001	In the context of automatic feeling recognition, understanding the character of feeling isn't associate finish in itself. It matters primarily as a result of concepts regarding the character of feeling from the manner emotional states area unit represented. They imply that bound options associated relationships area unit relevant to describing an emotion, characteristic it from others, associated crucial whether or not it qualifies as an emotion in the least.

CHAPTER 3

OVERALL DESIGN FOR PROPOSED SYSTEM

3.1 Software Requirement

For the development of this project, the version of python used is 3.6.5

Hardware Interfaces:

1. Processor: The 5th gen of Intel-Core i5 and with min speed of 2.9 GHz.
2. RAM : The min requirement is 4gb.
3. Hard Disk : The minimum requirement is 250gb

Software Interfaces:

- i. MS Word(2003)
- ii. Data-base storage : MS Excel
- iii. Operating-system(OS) : Win-10

3.2 Planning

The various steps required when making this project are below:

1. Research of problem statement.
2. Collection of the set requirements
3. Project feasibility analysis.Koolagudi and Rao (2012)
4. Development of a standard structure.

5. Journals about past activities related to this area.
6. To choose the best method for improving the algorithm.
7. We had various good and bad analyzes.
8. Initiate project development
9. Installation of software and PIP packages.
10. Improving algorithm.
11. Manual algorithm analysis.
12. The code should be according to the advanced PYTHON to enhance accuracy.

The SDLC used in our project is the iterative waterfall model

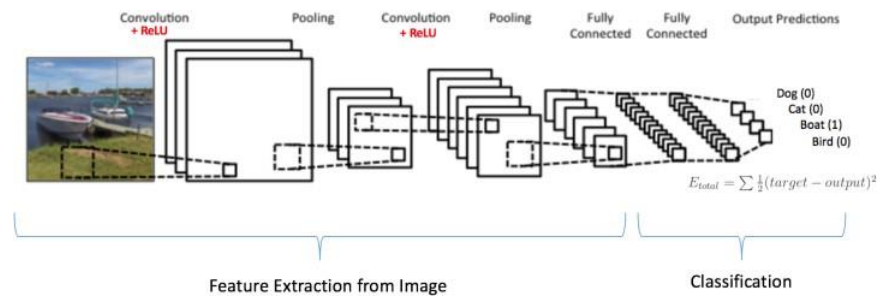


Figure 3.1: The Division of layers throughout

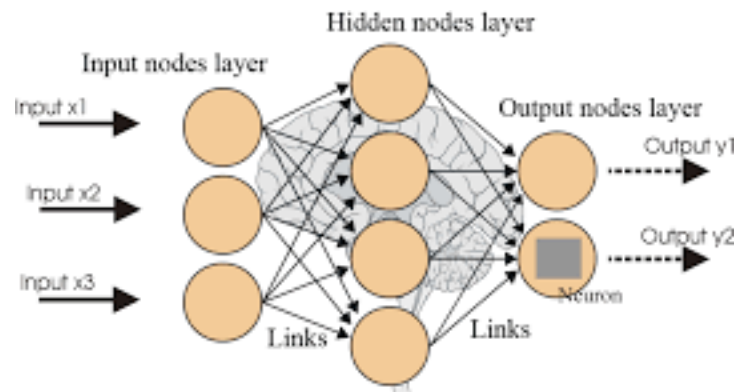


Figure 3.2: Artificial Neural Network

3.3 UML

Business analysts, software architects, and developers use UML to define, specify, create, and record current or new venture process, as well as the arrangement and manner of software system artefacts. UML is used in a variety of applications. It's compatible with all crucial object and pieces of program development methodologies, as well as a variety of implementation platforms. When it comes to modelling them with Machine Learning, the technique you choose has a big impact. When a category model of human emotion is selected, a classifier is almost certainly built, and texts and images are labelled with human emotions such as "happy," "sad," and so on. When using a dimensional model, however, outputs must be on a sliding scale.

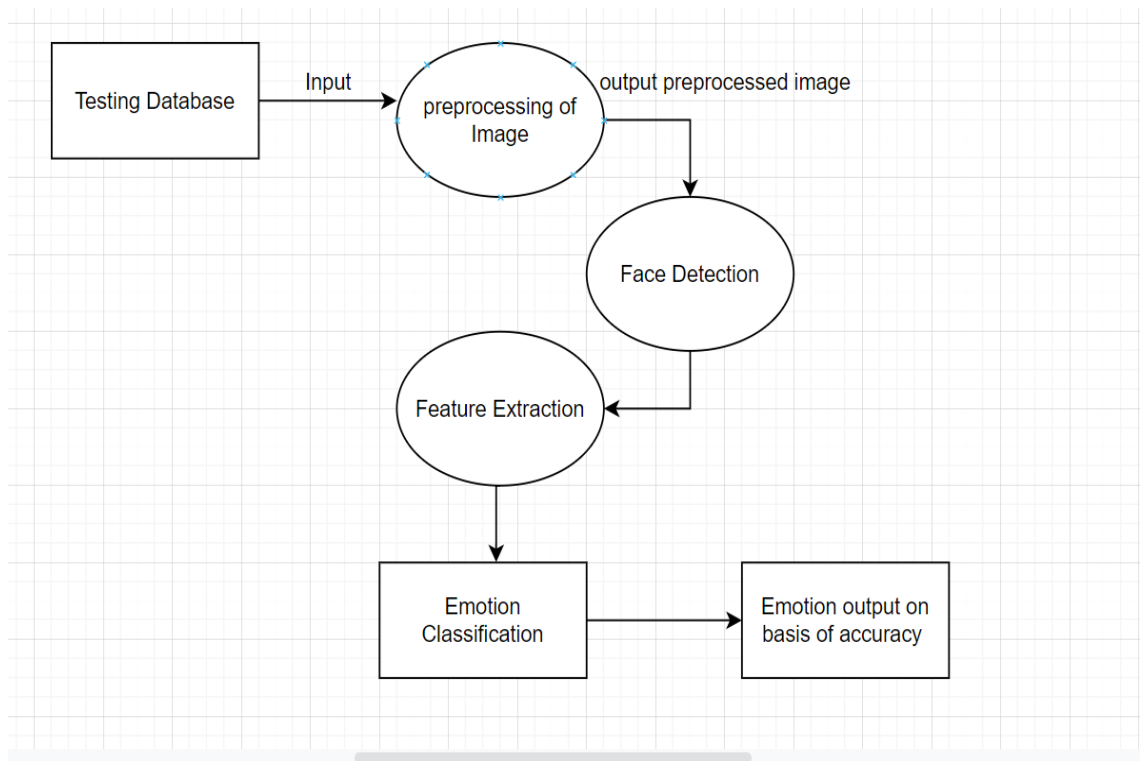


Figure 3.3: UML

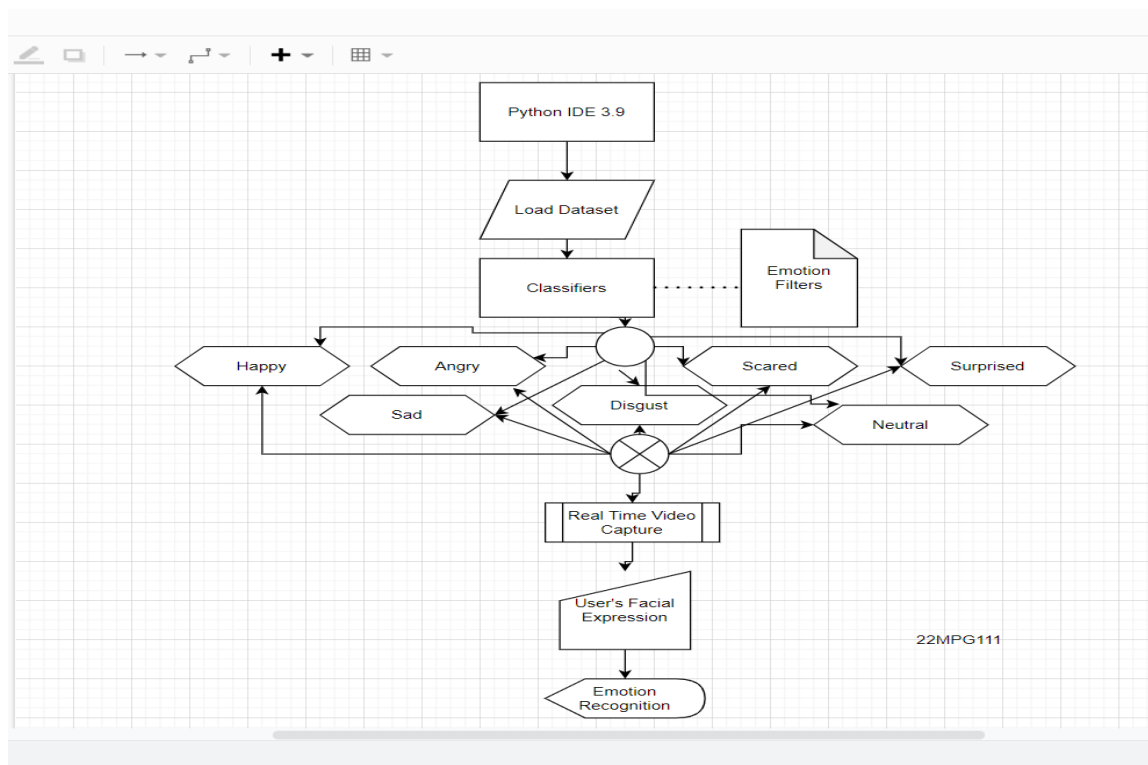


Figure 3.4: Emotion Classification

In the above figure: From the testing data, preprocessing of the image takes place and then passing through classifiers face detection is done, then we have filters which performs the feature extraction and the image is passed for classification between 7 basic emotions that are present in our dataset and the dominant one is generated as output. In this next image we can see the classifier and filters, how they are interconnected to each other and then through real time video capture face data is being collected for the recognition of emotion

3.4 Techniques to be used

- Face Registration
- Haar Features
- Integral Images
- Adaboost
- Cascading
- Haar Cascade Classifier in OpenCv
- Deep CNN
- Convolutional Layers
- Pooling Layers

3.5 Algorithms

Step one: assortment of image information set. (This time we have chosen to use the FER2013 database with 35887 48x48-pixel grey face pictures labeled with seven classes of expressions/emotions)

Step two :Pre-image process.Cowie et al. (2001)

Step three :Face detection in every icon.

Step four :The cut face is regenerate to grey pictures.

Step five : The pipe ensures that the full image will be inserted into the input layer as a program (1, 48, 48) numpy.

Step six :Similar numpy members move to Convolution-2d layer.

Step seven :Convolution layer generates the feature-map.

Step eight :A compilation methodology referred to as Max-Pooling-2d that have (2, 2) matrix on a feature-map that solely retains an utmost variety of pixels.Goldman and Sripada (2005)

Step nine :During coaching, Neural network Forward Propagation and back distribution is done at component values.

Step ten :The Soft-max operate explains and defines itself as the chance of each emotional category. The training model is ready pointing out the small print in doable emotional formation on the face.El Ayadi et al. (2011)

CHAPTER 4

DATASET/DATABASE SPECIFICATION

Three information sets area unit ordinarily employed in completely various steps in formation of the model.

- The training model was first worked on the coaching data-set, which is a group of instances, accustomed working specifications (for example, the weights of connections between neurons in artificial neural networks) Tripathi et al. (2019)
- A supervised learning approach is used to train the model (e.g., a neural internet or a naïve Thomas Bayes classifier) using the coaching dataset (e.g. gradient descent or random gradient descent). In practise, the coaching dataset often consists of pairs of input vectors and the matching answer vector or scalar, which is referred to as the goal. Ko (2018)
- The present model is applied to the coaching dataset and generates a result for each input vector, which is then compared to the target.
- The parameters of the model area unit were altered based on the results of the comparison and, as a consequence, the specific learning formula used. Each variable selection and parameter estimation will be included in the model fitting.
- Following that, the fitted model is used to predict the solutions for the observation in a second-dataset known as the validation-dataset.
- The validating-dataset offers an unbiased analysis of a model work on the coaching data-set whereas standardization the model's hyperparameters (for example eg. the amount of hidden units in an exceedingly convolution neural network). Tokuhisa et al. (2008)
- Validating-dataset is utilised for early ending regularisation: if the error on the validation dataset increases, cease coaching, as this is frequently an indicator of overfitting to the coaching dataset. This

simple approach is made more difficult to execute by the fact that the error in the validation dataset might change over coaching, resulting in several native minima. This complexity has resulted in the development of several ad hoc guidelines for determining whether overfitting has truly commenced. Kosti et al. (2017)

- At last now we, verify data-set could be the data-set accustomed giving unbiased analysis of the ending model matching the coaching dataset.
- the info we tend to use comprise of (48x48) picture element gray-scale picture profiles.
- The profiles are mechanically listed in order that the face is additional or a little bit focused and holds concerning constant quantity of area in every image. Schuller et al. (2003)
- The duty is to classify every profile supported in the feeling depicted within the face features into 1 in every 7 classes (Angry, Happy, Sad, Disgust, Fear, Surprise, Neutral).
- The coaching dataset contains 28709 eg, therefore validating set contains 3589 eg.

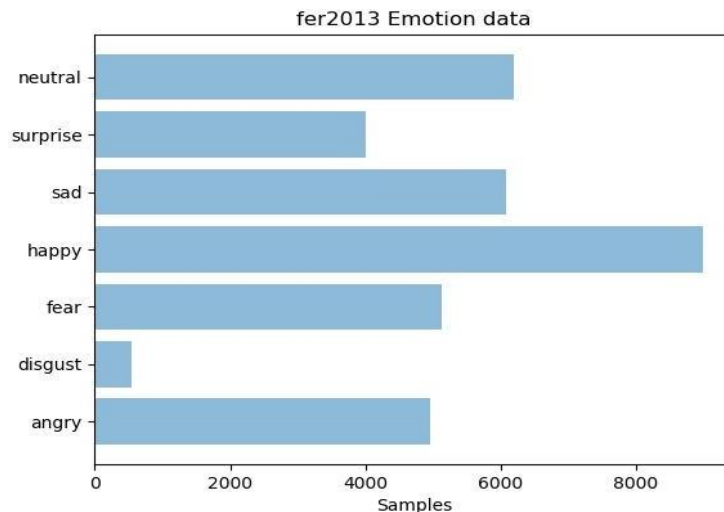


Figure 4.1: Emotion Dataset

Table 4.1: Emotion Dataset Table

Sno.	No. of images	Emotion
1	4593	Angry
2	547	Disgust
3	5121	Fear
4	8989	Happy
5	6077	Sad
6	4002	Surprise
7	6198	Neutral



Figure 4.2: Emotion Labelled data-set

4.1 Library and Packages

- Open-CV
- Numpy
- Numpy Array
- SciPy
- Keras
- TensorFlow

CHAPTER 5

METHODOLOGY

- **SYS**

Parameters and functions that are unique to the system. This module gives you access to some of the interpreter's variables and functions that act powerfully within interpreter. Using `help(SYS)` offers valuable detail info. Mower et al. (2010)

- **Sigmoid-Function :**

The sigmoid-function in an exceedingly neural-network can produce the the weights of inputs enhanced their top purpose (activation).

- **Softmax-Function :**

The Softmax-function calculate chances of sharing the event over n totally different events. generally method of claiming, this perform can calculate the chances of every target category over all attainable target categories.

- The important advantage of Softmax is that the output possibilities vary. The vary can zero to one, and also the total of all the chances are going to be capable one. Within soft-max function, it returns the odds of each category and also the target category might have a high chance when employed in a supervised multi-classification model.
- The formula showcase exponential (e-power) in the given input profile worth thus, also the total of exponential values within the inputs. Then quantitative relation depicts exponential of input worth and also the total of exponential results provides outcome of the softmax-function.

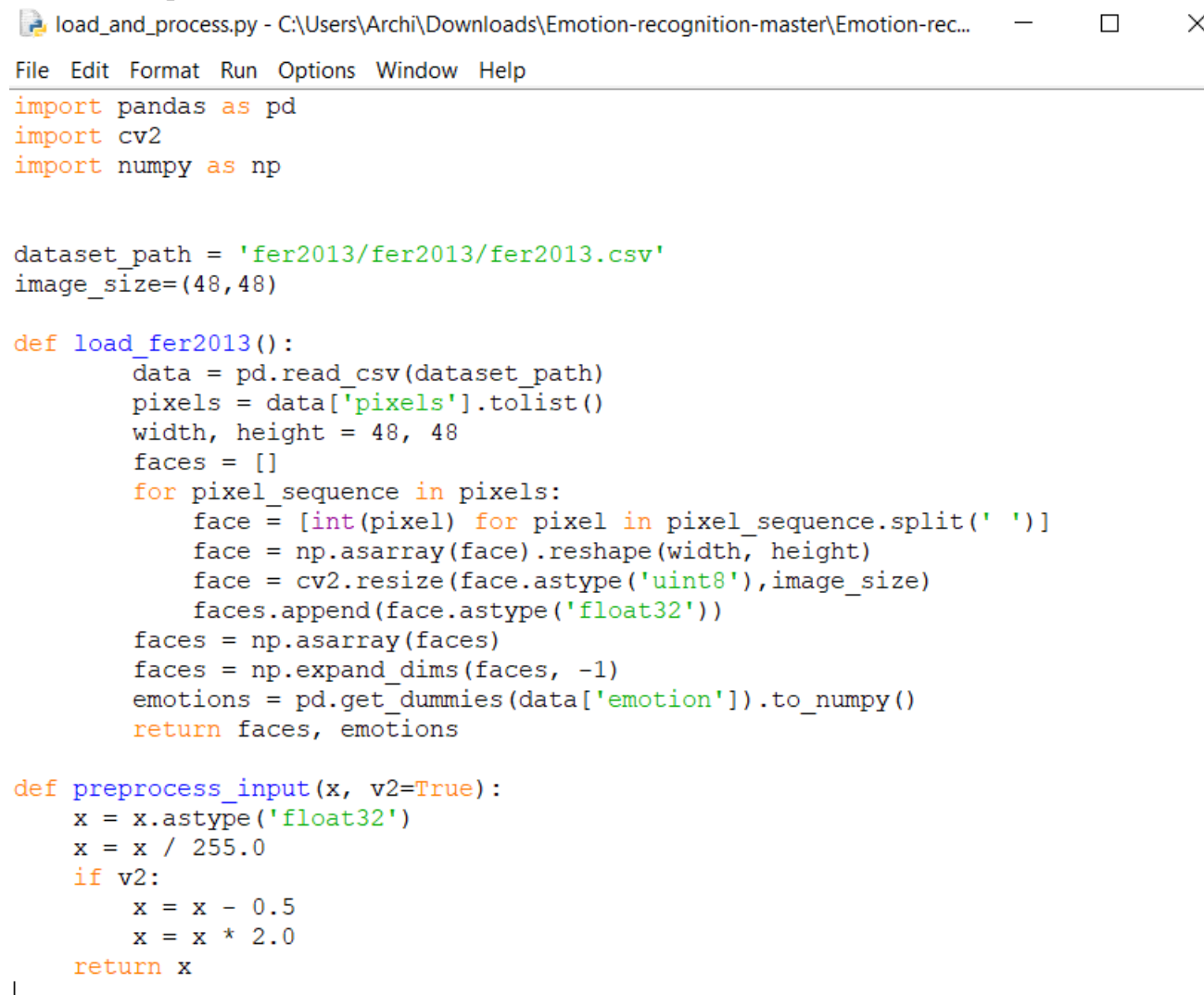
CHAPTER 6

CODING, TESTING

6.1 EXPERIMENTAL RESULTS

Three files used for the code are:

1. load and process



```
load_and_process.py - C:\Users\Archi\Downloads\Emotion-recognition-master\Emotion-rec...
File Edit Format Run Options Window Help
import pandas as pd
import cv2
import numpy as np

dataset_path = 'fer2013/fer2013/fer2013.csv'
image_size=(48,48)

def load_fer2013():
    data = pd.read_csv(dataset_path)
    pixels = data['pixels'].tolist()
    width, height = 48, 48
    faces = []
    for pixel_sequence in pixels:
        face = [int(pixel) for pixel in pixel_sequence.split(' ')]
        face = np.asarray(face).reshape(width, height)
        face = cv2.resize(face.astype('uint8'), image_size)
        faces.append(face.astype('float32'))
    faces = np.asarray(faces)
    faces = np.expand_dims(faces, -1)
    emotions = pd.get_dummies(data['emotion']).to_numpy()
    return faces, emotions

def preprocess_input(x, v2=True):
    x = x.astype('float32')
    x = x / 255.0
    if v2:
        x = x - 0.5
        x = x * 2.0
    return x
```

2.train emotion classifier

 train_emotion_classifier.py - C:\Users\Archi\Downloads\Emotion-recognition-master\Emotion-recognition-mast...

File Edit Format Run Options Window Help

```
"""
Description: Train emotion classification model
"""

from keras.callbacks import CSVLogger, ModelCheckpoint, EarlyStopping
from keras.callbacks import ReduceLROnPlateau
from keras.preprocessing.image import ImageDataGenerator
from load_and_process import load_fer2013
from load_and_process import preprocess_input
from models.cnn import mini_XCEPTION
from sklearn.model_selection import train_test_split

# parameters
batch_size = 32
num_epochs = 10
input_shape = (48, 48, 1)
validation_split = .2
verbose = 1
num_classes = 7
patience = 50
base_path = 'models/'

# data generator
data_generator = ImageDataGenerator(
    featurewise_center=False,
    featurewise_std_normalization=False,
    rotation_range=10,
    width_shift_range=0.1,
    height_shift_range=0.1,
    zoom_range=.1,
    horizontal_flip=True)

# model parameters/compilation
model = mini_XCEPTION(input_shape, num_classes)
model.compile(optimizer='adam', loss='categorical_crossentropy',
              metrics=['accuracy'])
model.summary()
```

```

# callbacks
log_file_path = base_path + '_emotion_training.log'
csv_logger = CSVLogger(log_file_path, append=False)
early_stop = EarlyStopping('val_loss', patience=patience)
reduce_lr = ReduceLROnPlateau('val_loss', factor=0.1,
                               patience=int(patience/4), verbose=1)

trained_models_path = base_path + '_mini_XCEPTION'
model_names = trained_models_path + '.{epoch:02d}-{val_accuracy:.2f}.hdf5'
model_checkpoint = ModelCheckpoint(model_names, 'val_loss', verbose=1,
                                   save_best_only=True)
callbacks = [model_checkpoint, csv_logger, early_stop, reduce_lr]

# loading dataset
faces, emotions = load_fer2013() # using csv file here
faces = preprocess_input(faces)
num_samples, num_classes = emotions.shape
xtrain, xtest, ytrain, ytest = train_test_split(faces, emotions, test_size=0.2, shuffle=True)
model.fit_generator(data_generator.flow(xtrain, ytrain,
                                       batch_size,
                                       steps_per_epoch=len(xtrain) / batch_size,
                                       epochs=num_epochs, verbose=1, callbacks=callbacks,
                                       validation_data=(xtest, ytest))

```

3. real time video

```

real_time_video.py - C:\Users\Arch\Downloads\Emotion-recognition-master\Emotion-recognition-master\real_time_video.py (3.10.2)
File Edit Format Run Options Window Help

from keras.preprocessing.image import img_to_array
import cv2
from keras.models import load_model
import numpy as np

# parameters for loading data and images
detection_model_path = 'haarcascade_files/haarcascade_frontalface_default.xml'
emotion_model_path = 'models/_mini_XCEPTION.102-0.66.hdf5'

# hyper-parameters for bounding boxes shape
# loading models
face_detection = cv2.CascadeClassifier(detection_model_path)
emotion_classifier = load_model(emotion_model_path, compile=False)
EMOTIONS = ["angry", "disgust", "scared", "happy", "sad", "surprised",
            "neutral"]

# feelings_faces = []
# for index, emotion in enumerate(EMOTIONS):
#     feelings_faces.append(cv2.imread('emojis/' + emotion + '.png', -1))

# starting video streaming
cv2.namedWindow('your_face')
camera = cv2.VideoCapture(0)
while True:
    frame = camera.read()[1]
    # reading the frame
    height, width, depth = frame.shape
    imgScale = 300 / width
    frame = cv2.resize(frame, None, fx=imgScale, fy=imgScale)
    gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
    faces = face_detection.detectMultiScale(gray, scaleFactor=1.1, minNeighbors=5, minSize=(30, 30), flags=cv2.CASCADE_SCALE_IMAGE)

    canvas = np.zeros((250, 300, 3), dtype="uint8")
    frameClone = frame.copy()
    if len(faces) > 0:
        faces = sorted(faces, reverse=True,
                      key=lambda x: (x[2] - x[0]) * (x[3] - x[1]))[0]
        (fX, fY, fW, fH) = faces
        # Extract the ROI of the face from the grayscale image, resize it to a fixed 28x28 pixels, and then prepare
        # the ROI for classification via the CNN
        roi = gray[fY:fY + fH, fX:fX + fW]
        roi = cv2.resize(roi, (64, 64))
        roi = roi.astype("float") / 255.0
        roi = img_to_array(roi)
        roi = np.expand_dims(roi, axis=0)

```

```

        preds = emotion_classifier.predict(roi)[0]
        emotion_probability = np.max(preds)
        label = EMOTIONS[preds.argmax()]
    else: continue

    for (i, (emotion, prob)) in enumerate(zip(EMOTIONS, preds)):
        # construct the label text
        text = "{}: {:.2f}%".format(emotion, prob * 100)

        # draw the label + probability bar on the canvas
        # emoji_face = feelings_faces[np.argmax(preds)]

        w = int(prob * 300)
        cv2.rectangle(canvas, (7, (i * 35) + 5),
            (w, (i * 35) + 35), (0, 0, 255), -1)
        cv2.putText(canvas, text, (10, (i * 35) + 23),
            cv2.FONT_HERSHEY_SIMPLEX, 0.45,
            (255, 255, 255), 2)
        cv2.putText(frameClone, label, (fx, fy - 10),
            cv2.FONT_HERSHEY_SIMPLEX, 0.45, (0, 0, 255), 2)
        cv2.rectangle(frameClone, (fx, fy), (fx + fw, fy + fh),
            (0, 0, 255), 2)

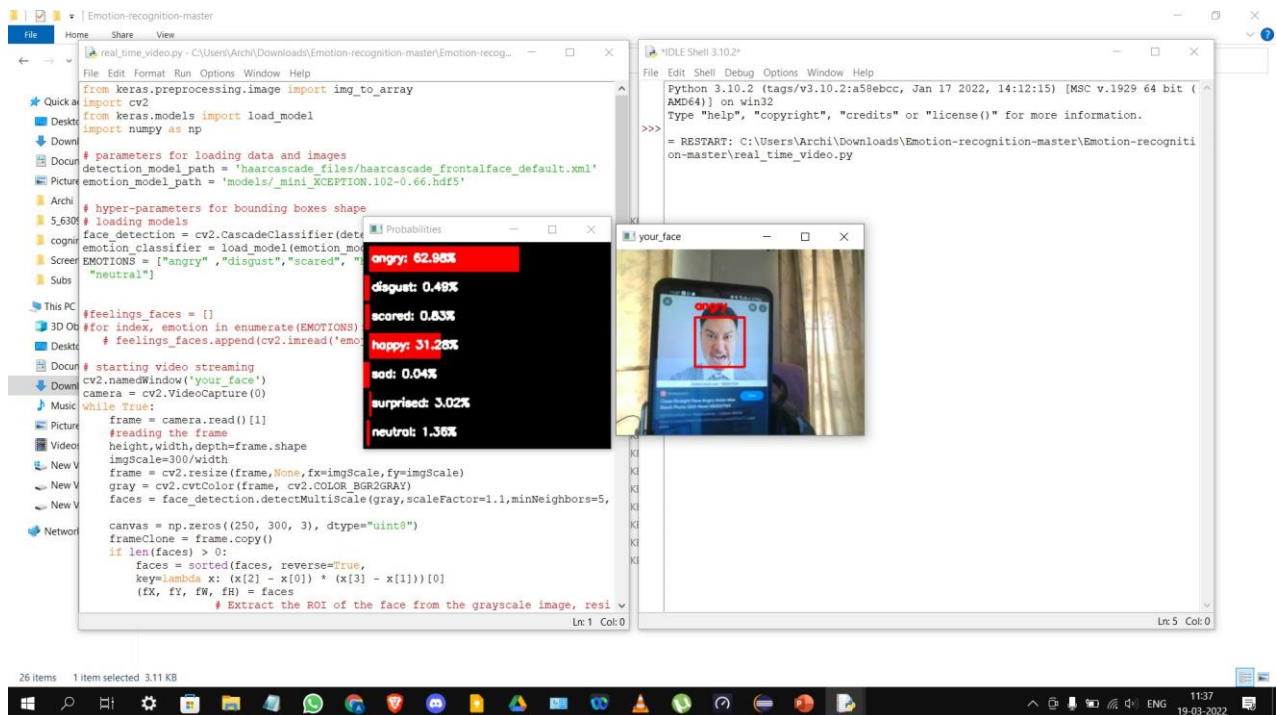
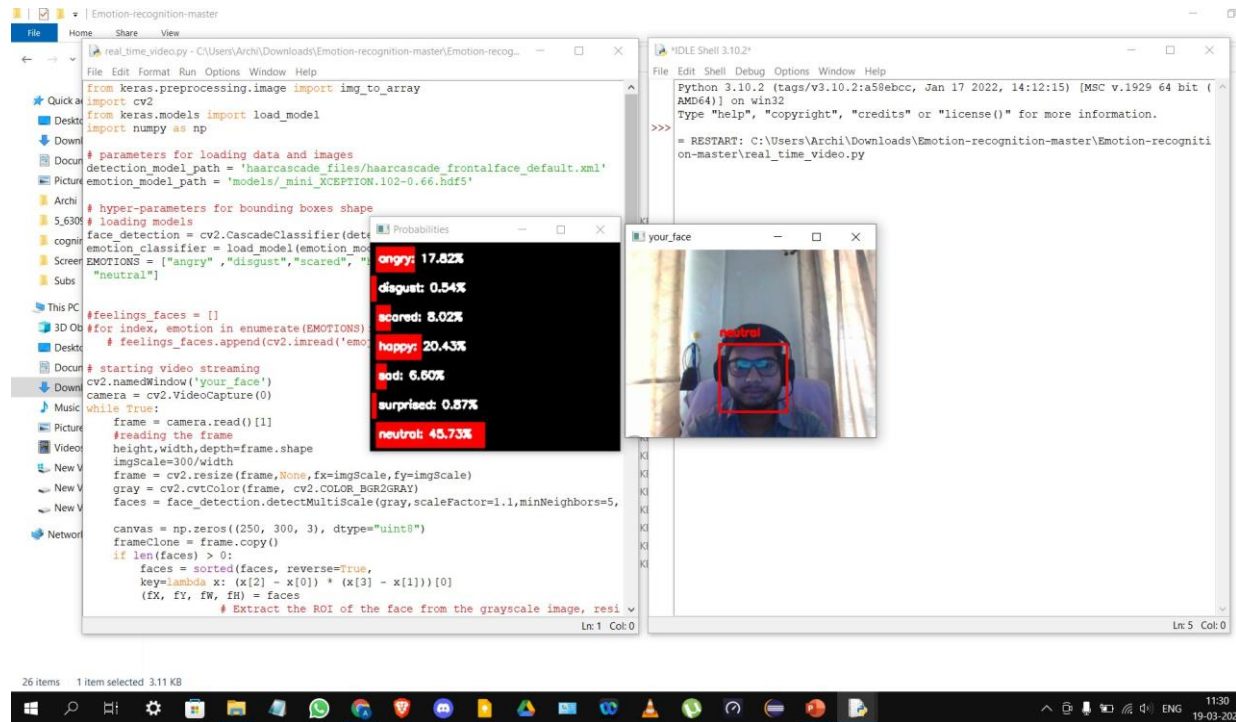
    # for c in range(0, 3):
    #     frame[200:320, 10:130, c] = emoji_face[:, :, c] * \
    #         (emoji_face[:, :, 3] / 255.0) + frame[200:320,
    #         10:130, c] * (1.0 - emoji_face[:, :, 3] / 255.0)

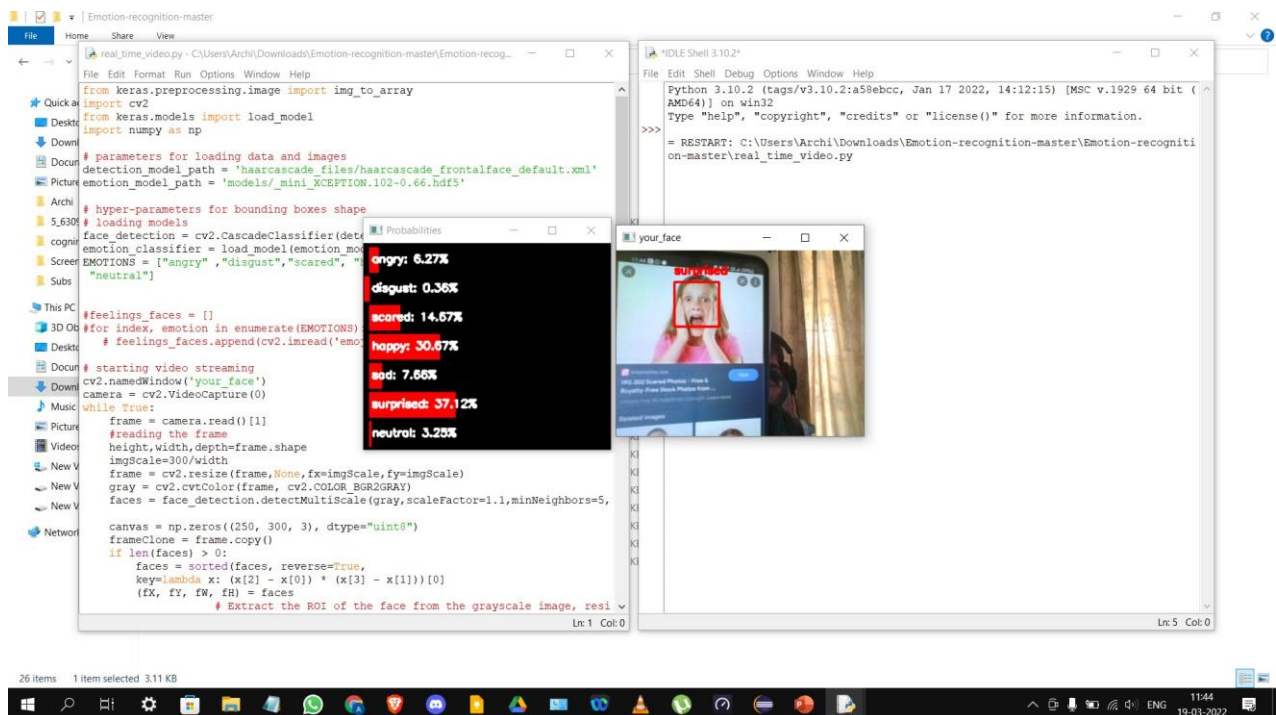
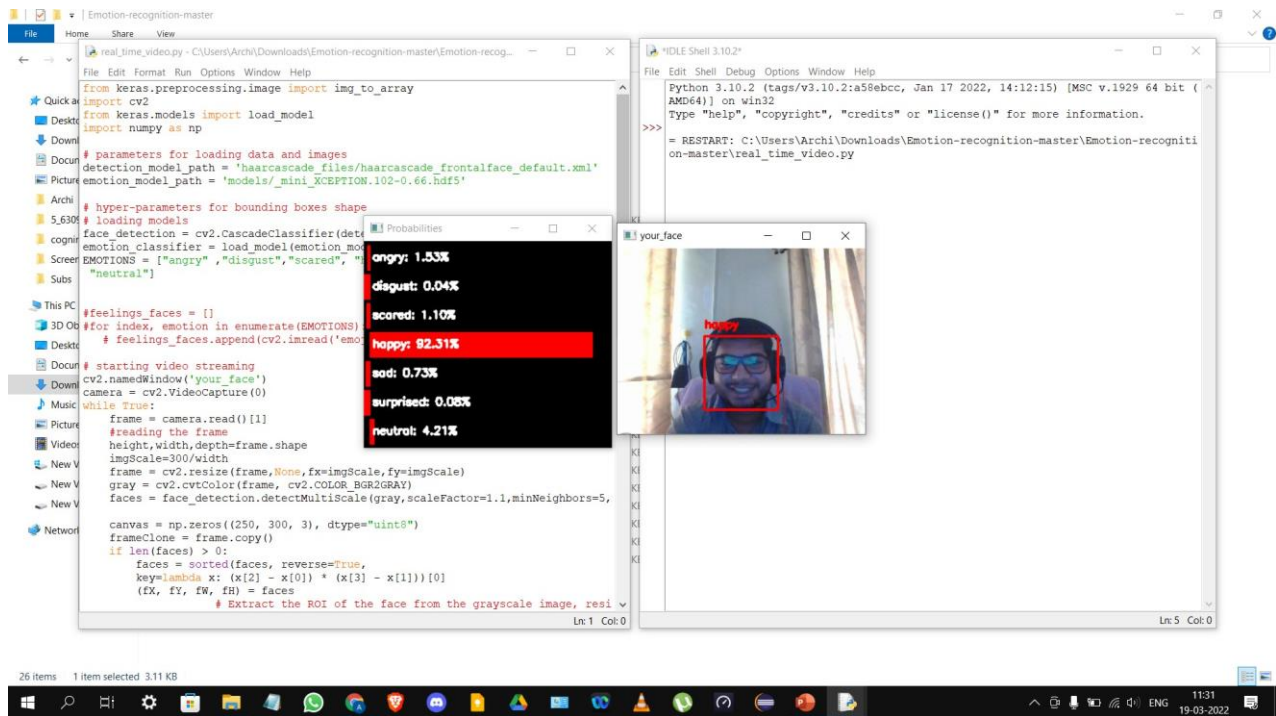
    cv2.imshow('your_face', frameClone)
    cv2.imshow("Probabilities", canvas)
    if cv2.waitKey(1) & 0xFF == ord('q'):
        break

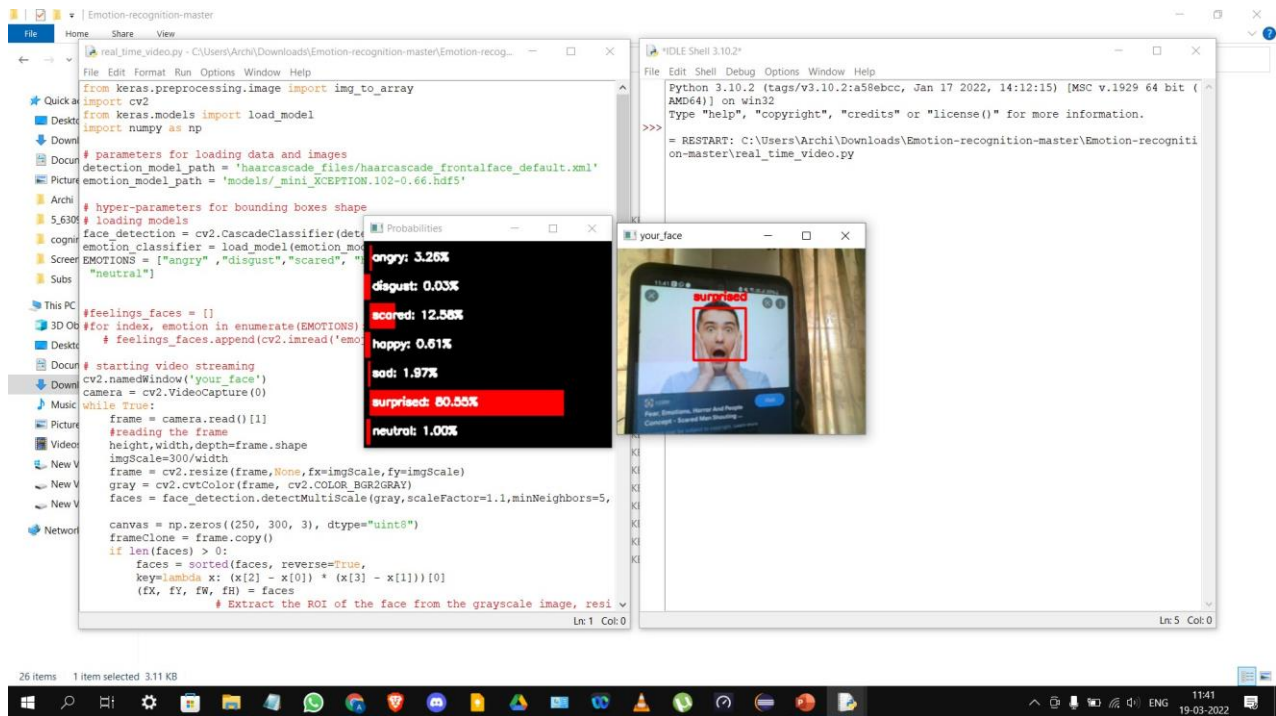
camera.release()
cv2.destroyAllWindows()

```

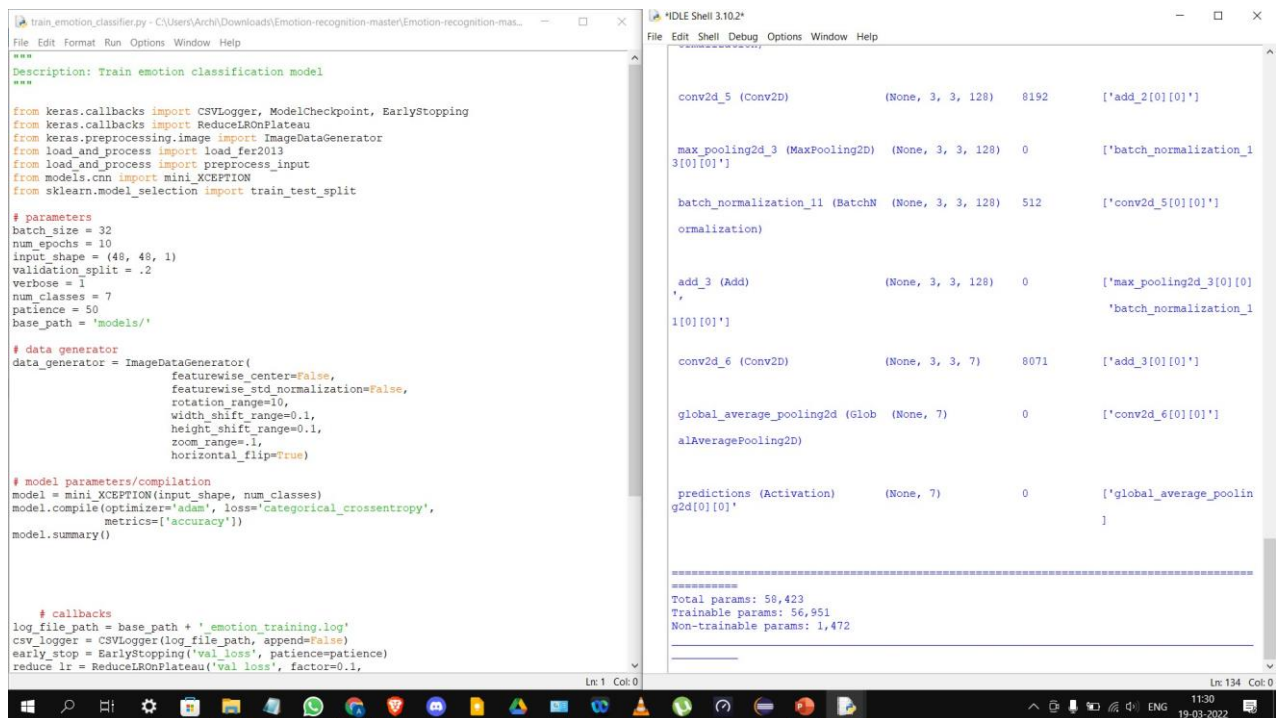
The following emotions are the output as given below:







6.2 PERFORMANCE EVALUATION



```

Python 3.10.2 (tags/v3.10.2:a58ebcc, Jan 17 2022, 14:12:15) [MSC v.1929 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\Users\Archi\Downloads\Emotion-recognition-master\Emotion-recognition-master\train_emotion_classifier.py
Model: "model"

```

Layer (type)	Output Shape	Param #	Connected to
input_1 (InputLayer)	(None, 48, 48, 1)	0	[]
conv2d (Conv2D)	(None, 46, 46, 8)	72	['input_1[0][0]']
batch_normalization (Batch Normalization)	(None, 46, 46, 8)	32	['conv2d[0][0]']
activation (Activation)	(None, 46, 46, 8)	0	['batch_normalization[0][0]']
conv2d_1 (Conv2D)	(None, 44, 44, 8)	576	['activation[0][0]']
batch_normalization_1 (Batch Normalization)	(None, 44, 44, 8)	32	['conv2d_1[0][0]']
activation_1 (Activation)	(None, 44, 44, 8)	0	['batch_normalization_1[0][0]']
separable_conv2d (Separable Conv2D)	(None, 44, 44, 16)	200	['activation_1[0][0]']
batch_normalization_3 (Batch Normalization)	(None, 44, 44, 16)	64	['separable_conv2d[0][0]']
activation_2 (Activation)	(None, 44, 44, 16)	0	['batch_normalization_3[0][0]']
separable_conv2d_1 (Separable Conv2D)	(None, 44, 44, 16)	400	['activation_2[0][0]']
batch_normalization_4 (Batch Normalization)	(None, 44, 44, 16)	64	['separable_conv2d_1[0][0]']
conv2d_2 (Conv2D)	(None, 22, 22, 16)	128	['activation_1[0][0]']
max_pooling2d (Max Pooling2D)	(None, 22, 22, 16)	0	['batch_normalization_4[0][0]']
batch_normalization_2 (Batch Normalization)	(None, 22, 22, 16)	64	['conv2d_2[0][0]']
add (Add)	(None, 22, 22, 16)	0	['max_pooling2d[0][0]', 'batch_normalization_2[0][0]']

```

max_pooling2d_2 (Max Pooling2D) (None, 6, 6, 64) 0 ['batch_normalization_10[0][0]']
batch_normalization_8 (Batch Normalization) (None, 6, 6, 64) 256 ['conv2d_4[0][0]']
add_2 (Add) (None, 6, 6, 64) 0 ['max_pooling2d_2[0][0]', 'batch_normalization_8[0][0]']
separable_conv2d_6 (Separable Conv2D) (None, 6, 6, 128) 8768 ['add_2[0][0]']
batch_normalization_12 (Batch Normalization) (None, 6, 6, 128) 512 ['separable_conv2d_6[0][0]']
activation_5 (Activation) (None, 6, 6, 128) 0 ['batch_normalization_12[0][0]']
separable_conv2d_7 (Separable Conv2D) (None, 6, 6, 128) 17536 ['activation_5[0][0]']
batch_normalization_13 (Batch Normalization) (None, 6, 6, 128) 512 ['separable_conv2d_7[0][0]']
conv2d_5 (Conv2D) (None, 3, 3, 128) 8192 ['add_2[0][0]']
max_pooling2d_3 (Max Pooling2D) (None, 3, 3, 128) 0 ['batch_normalization_13[0][0]']
batch_normalization_11 (Batch Normalization) (None, 3, 3, 128) 512 ['conv2d_5[0][0]']
add_3 (Add) (None, 3, 3, 128) 0 ['max_pooling2d_3[0][0]', 'batch_normalization_11[0][0]']
conv2d_6 (Conv2D) (None, 3, 3, 7) 8071 ['add_3[0][0]']
global_average_pooling2d (Global Average Pooling2D) (None, 7) 0 ['conv2d_6[0][0]']
predictions (Activation) (None, 7) 0 ['global_average_pooling2d[0][0]']

```

Total params: 58,423
 Trainable params: 56,951
 Non-trainable params: 1,472

CHAPTER 7

COMPARISON WITH EXISTING SYSTEM

Table 7.1: Comparison Table

Comparison Chart: Market vs Our Model	
There are different models present in the market for emotion classification that uses graph neural network or capsule neural network	Our Approach: We used the convolutional neural network (CNN) compared to its predecessors in that it mechanically detects the necessary options while not any human oversight
Humans try to perceive and understand the emotion of others through their mental filters and feelings	Emotion Recognition System: It is achieved by the capacity to read, understand and learn about the emotional life of humans
BAUM is a video-based dataset. The dataset contains audio-visual clips of various languages that are annotated. The clips represent real-world events like totally different poses, lighting conditions, and subjects of varying ages. associate degree image-based dataset is created with the heightframes from every clip. The BAUM-2i is an associate degree free expression database, whose samples are collected from movies and television series. 998 pictures labeled with seven expressions, selected for analysis.	FER-2013 is a well-studied dataset. It is one among the more difficult datasets with human-level accuracy only at $65\pm5\%$ and the best playacting published works achieving seventy-five.2% test accuracy. simply downloadable on Kaggle, the dataset has 35,887 contained images that are normalized to 48x48 pixels in grayscale. It contains seven facial expressions, with distributions of Angry (4,953), Disgust (547), Fear (5,121), Happy (8,989), Sad (6,077), Surprise (4,002), and Neutral (6,198)

CHAPTER 8

CONCLUSION

8.1 Key Findings

Expression is usually a mixture of 2 or a lot of of the archetypal expressions. Also, expression is supposed to be particular and to start and finish with a neutral position. In reality, facial expression is far more advanced and occur in numerous combos and intensity. Thus, user known expressions can be a mixture of 2 totally different expressions with one in all them being a lot of main strength. The supervised classification, therefore, ought to be good enough to properly determine the mix of expressions and every emotion's depth.

8.2 Significance

Broader applications: The output of a neural-network rests on the sort of variables extracted from the facial expression in the picture. It is wide applied to varied analysis areas, like mental diseases designation and human social/physiological interaction detection.

8.3 Limitations

- Different types and versions of software have drawbacks, such as dataset input being limited to textual data and image.
- The accuracy of the sensors utilized in the emotion-sensing system, such as cameras, thermal image sensors, and the facial recognition algorithm used, determines the system's performance and outcomes. Due to the utilization of expensive components, a highly accurate system will be costly.

8.4 Future Scope

We show a framework of an automatic Emotion Recognition system aided by multimodal sensor data, also as a theoretical analysis of its practicableness and possibleness for feeling detection; its utility can possibly be shown within the future through real-world trials.

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APPENDIX A

PLAGIARISM REPORT

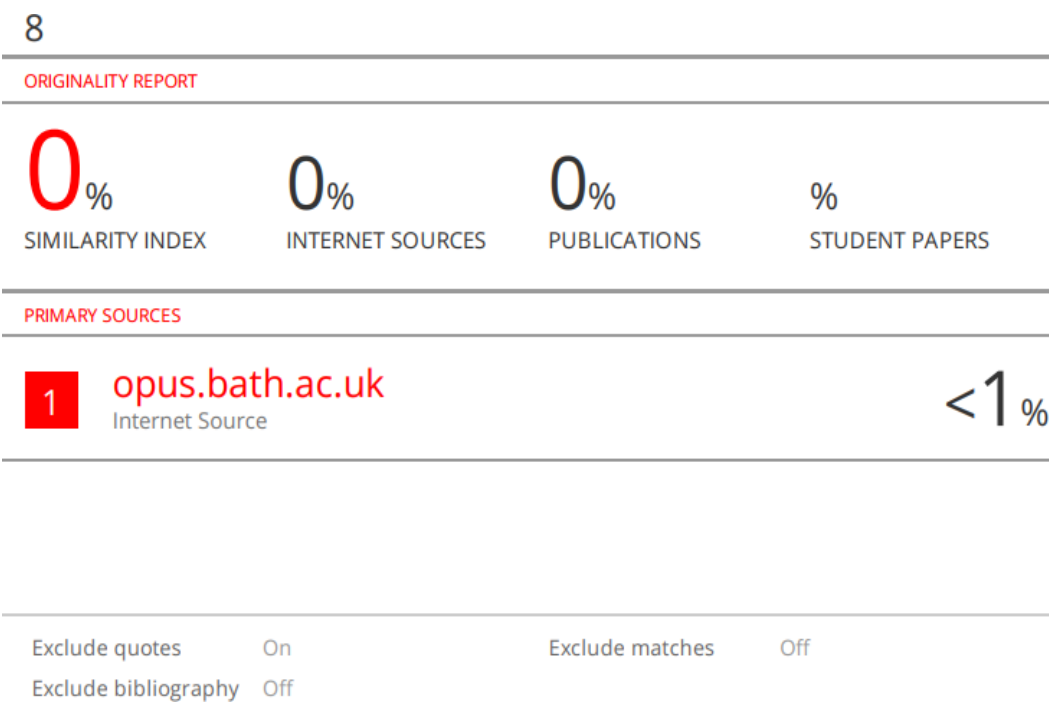


Figure A.1: Plagiarism Report